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Summary

Senior Machine Learning Engineer with **6+** years of experience specializing in LLM training, reinforcement learning, and agentic AI systems. Experienced in building and deploying production-grade ML systems at enterprise scale. Published researcher with work accepted at **tier-1 AI conferences** and patents to credit.

Publications

- [1] **EnterpriseOps-Gym: Environments and Evaluations for Stateful Agentic Planning and Tool Use in Enterprise Settings**
arXiv preprint, March 2026 arXiv:2603.13594
- [2] **Apriel-Nemotron-15B-Thinker**
arXiv preprint, August 2025 arXiv:2508.10948
- [3] **DNR Bench: Benchmarking Over-Reasoning in Reasoning LLMs**
AAAI 2026 (Oral), March 2025 arXiv:2503.15793

Experience

ServiceNow

March 2024 – Present

Senior Machine Learning Engineer

Hyderabad, India

- **Reinforcement Learning for Agents:** Optimized large language models for agentic applications using advanced RL algorithms (GRPO, DAPO, PPO), achieving improved decision-making and task efficiency.
- **Agentic Orchestration:** Designed and implemented a multi-agent orchestration framework with **LangGraph** to coordinate complex tasks and automate enterprise-scale workflows.
- **Synthetic Data Generation Framework:** Built a graph-based framework leveraging **LangGraph** to generate high-quality synthetic datasets for Supervised Fine-Tuning (SFT) and Reinforcement Learning pipelines.
- **Reasoning Benchmark:** Developed a standardised benchmark to evaluate reasoning capabilities of LLMs across diverse tasks and scenarios.
- **Evaluation Framework:** Designed and developed a holistic evaluation framework using LangChain to analyse various capabilities and traits of LLMs by incorporating industry-standard datasets and metrics.

ServiceNow

September 2022 – February 2024

Machine Learning Engineer

Hyderabad, India

- **LLM Fine-tuning:** Fine-tuned and evaluated large language models like **StarCoder**, **StarCoderPlus**, **Llama 2** to perform tasks like text summarization, multi-turn Q&A and code generation specific to ServiceNow.
- **Inference Pipeline:** Developed an inference pipeline for large language models (LLMs) using **vLLM**, **TGI**, and **TensorRT**, incorporating model quantization and continuous batching
- **Similarity Service:** Developed and architected an enterprise-level **Similarity Search** service using the state-of-the-art Transformer-based models, **ONNX**, **HNSW** and integrated NLP algorithms for accurate similarity analysis.
- **Clustering Service:** Built an efficient **KMeans Clustering** Service with prefiltering capabilities using **FAISS**, enabling hassle-free clustering of large datasets at an enterprise level.
- **Mentorship:** Provided mentorship and guidance to **two interns**, resulting in the successful completion of their internship projects.

ServiceNow

March 2022 – August 2022

Software Engineer - AI/ML

Hyderabad, India

- **ML Automation:** Developed and deployed ML-based automation to identify customer issues that could be resolved by ChatBot, deflecting **1.75M USD** worth of tickets from human resolution every quarter.

ServiceNow

July 2020 – February 2022

Associate Software Engineer - AI/ML

Hyderabad, India

- **Semantic Search:** Built and deployed semantic search service using the RoBERTa algorithm to retrieve relevant articles for customer incident tickets, resolving **1.35M USD** worth of tickets per year without human intervention.
- **Virtual Agent:** Created and maintained virtual agent topics using NLU (Natural Language Understanding), thereby improving customer experience and reducing costs.

- **Contract Analytics:** Developed and deployed NLP models to extract relevant features from employment contracts using custom datasets and advanced NLP algorithms, such as **RNN**, **LSTM**, and **RASA NLU**.

Technical Skills

Languages: Python, SQL, JavaScript, Java
ML & RL: PPO, GRPO, DAPO, SFT, DPO, RLHF, BERT, RoBERTa, Transformers, Diffusion Models, RNN, LSTM
LLM & Agentic: LangChain, LangGraph, vLLM, TensorRT, DeepSpeed, ONNX. TensorFlow, PyTorch
Evaluation & Benchmarking: LLM-as-Judge, Reward Modeling, Benchmark Design, Synthetic Data Generation
Infrastructure: FAISS, Docker, Git, Azure, H100/GPU Clusters, NCCL, Distributed Training, Django, Flask

Education

Government Model Engineering College, Kochi
Bachelor of Technology in Computer Science and Engineering
CGPA : 8.8/10

2016 – 2020
Kochi, India

Patents

Smart Persistence Of Model For Effective Predictions And Updates
US Patent No. 18/626,139

January 2026
Patent Issued

Platform Agnostic Scalable And High-performance Semantic Search Framework
US Patent No. 18/623,844

September 2025
Patent Issued

Framework Neutral And Updatable Clustering Model
US Patent Application No. 18/426,810

April 2024
Patent Filed

Awards and Recognition

ATG Llama Award: Recognized as a top contributor to the field of machine learning by ServiceNow, receiving the prestigious Llama Award **twice (2022 and 2023)** with a recognition ratio of just **14%**.